

TASK 1 – CRUD REST API USING SPRING BOOT

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Project Title

Employee Management System – CRUD REST API using Spring Boot

Objective

To develop a RESTful web service using Spring Boot that performs basic CRUD (Create, Read, Update, Delete) operations for managing employee data.

Tech Stack Used

- Java 17
- Spring Boot 3.x
- Spring Web
- Spring Data JPA
- H2 Database
- Lombok
- Thunder Client / Postman (for API testing)

Project Description

This project is a simple Employee Management System built using Spring Boot. It allows users to perform CRUD operations on employee data. The data is stored in an in-memory H2 database.

API Endpoints

Operation	Method	Endpoint
Create Employee	POST	/api/employees
Get All Employees	GET	/api/employees
Get Employee by ID	GET	/api/employees/{id}
Update Employee	PUT	/api/employees/{id}
Delete Employee	DELETE	/api/employees/{id}

Sample JSON (POST Request)

```
{  
  "name": "Keerthi",  
  "email": "keerthi@gmail.com",  
  "department": "IT"  
}
```

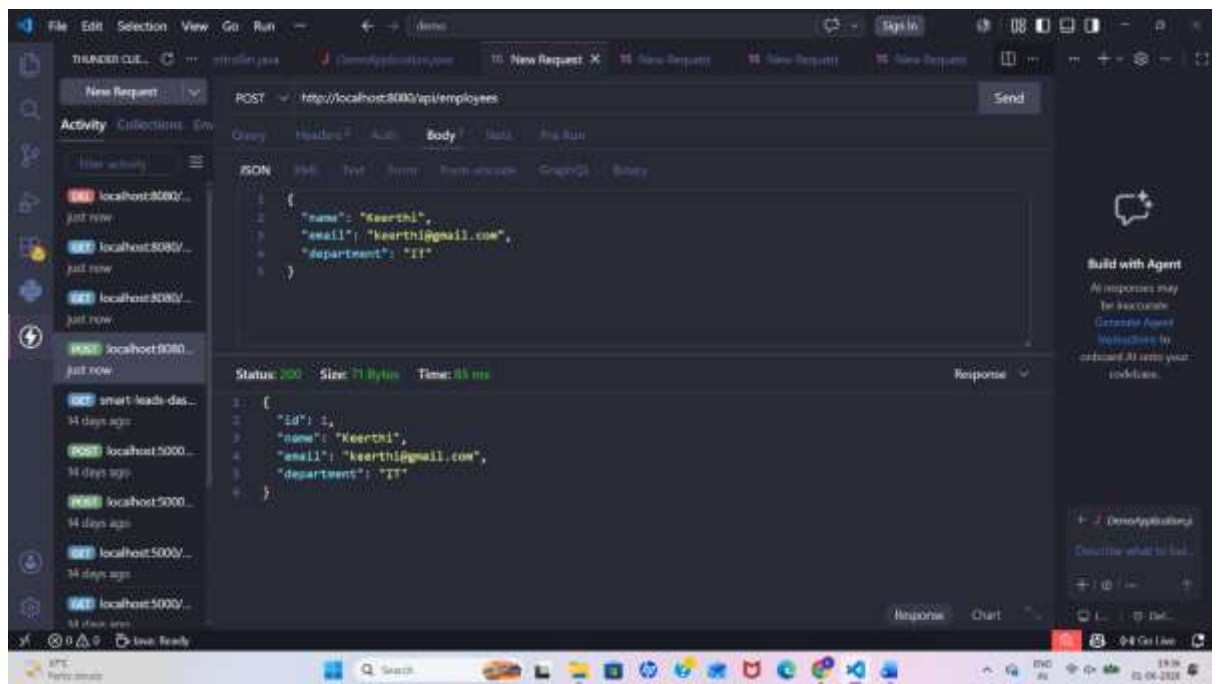
Testing Tool Used

- Thunder Client / Postman

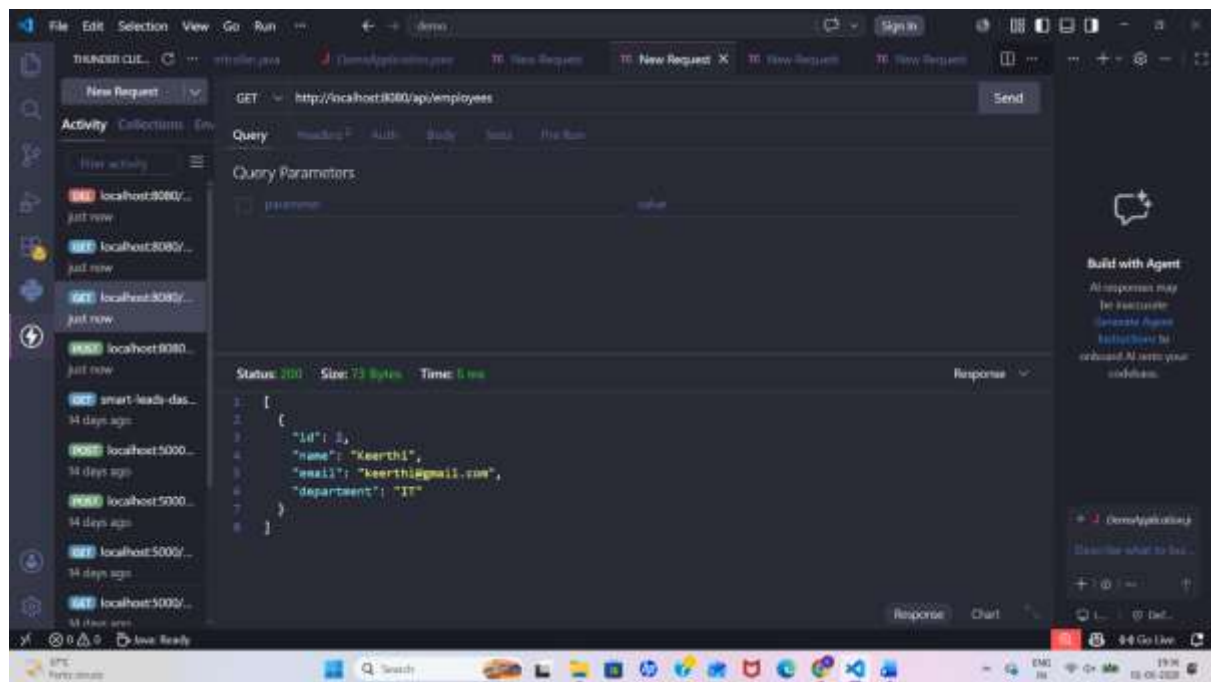
Screenshots

(Add here your screenshots of:)

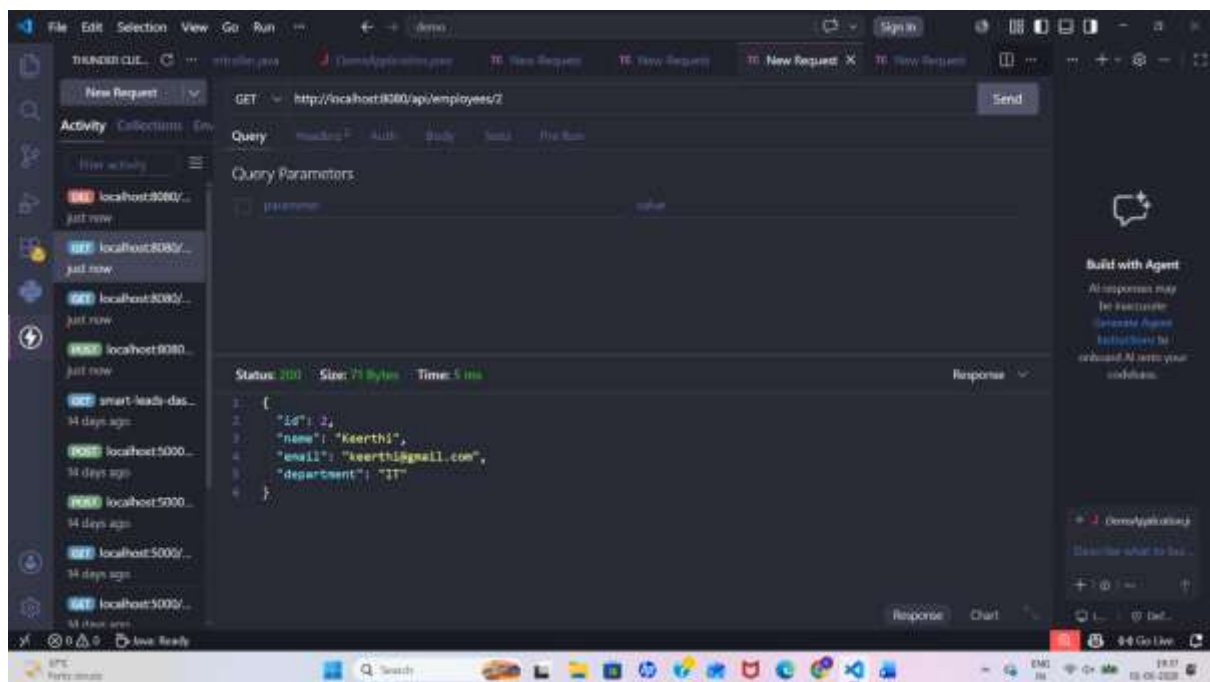
- POST request success



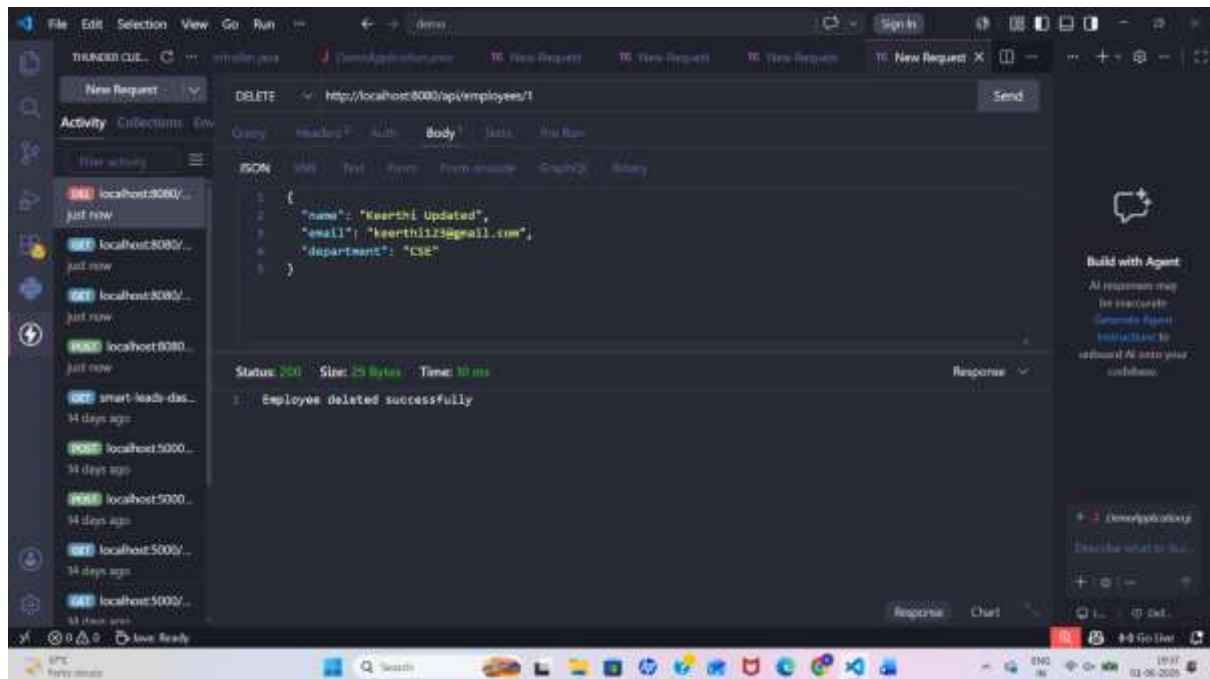
- GET all employees



- PUT update request



- DELETE request



Key Learnings

- Spring Boot REST API development
- CRUD operations using JPA
- Dependency injection and service layer architecture
- API testing using Thunder Client

GitHub Repository

Link: <https://github.com/keerthibandari167/employee-crud-api>

Conclusion

The Employee Management System using Spring Boot successfully demonstrates CRUD operations with REST API architecture and database integration.